Where the shipped chain stands and what is wrong (held-out test files of jobs 0-199, CC1p0pi selection, 2026-11-10)

Row 1: each block of the chain, what closes against Pandora and what is open. Row 2: the selection stage by stage. Row 3: the two open items traced to their cause. Green = closed or fixed, red = open.

Inputs + particle encoder
80 summary features + 64-d embedding of the post-FSI particle set (no Geant4).
closes
per-event direction of the generated tracks follows the event (corr. of dir_x with Pandora 0.57; 0.06 without the set); containment-flag heads AUC 0.965.
open
cardinality head trained on a table whose split differs: saw 90% of the held-out files (affects only whether a slice exists).

→

Slice generator (m2_official_m2vw)
topology + presence heads, one flow; vertex as residual; wall flags.
closes
50 column marginals (median W1 0.016); correlation matrix within 0.05; cohpi cut flow within 7% at every stage; vertex resolution 0.4 cm; wall pile-up matched.
open
exit <-> low-multiplicity link of CC Npi slices (n_pfp = 3, longest end in FV 0.498 vs 0.418): background +135 at containment.

→

PFP generator (m4_official_m3w0)
set flow over 12 PFP tokens x 38 fields; track end generated, wall flag per axis.
closes
field marginals (median W1 0.006); cohpi PFP counts; end on the wall 0.363 (0.349); per-event end-point distance 11.6 cm, at the two-draw floor of 14.5 cm.
open
cathode side of the ends not correlated: contained signal lost (0.866 vs 0.984), exiting-muon 1p0pi kept (293 vs 103).

→

Surrogated dataframes
surrogate_official_m2vw_pfp, jobs 0-199 (6 GB); 1,800 jobs pending review.
closes
production format; slc + pfp generated, truth tables copied; analyses run unchanged; test files = files no generator saw (fixed 2026-11-09).
missing
cosmic-only / unmatched slices; PFPs without a fit; induction-plane chi2; PFP truth-match columns.

→

CC1p0pi selection (M. Fernandez)
transcribed notebook cut flow; primary signal: 1mu1p0pi, p < 1 GeV, vertex in FV.
closes
every stage within 5% in selected events; final efficiency 0.113 (Pandora 0.109), final purity 0.869 (0.885).
open
2prong-contained purity 0.403 vs 0.425 (previous chain 0.398); truth-contained (notebook) purity 0.351 vs 0.418.

The selection stage by stage, shipped chain against Pandora on the same 180,000 held-out neutrinos (7,659 primary signal)

stage	Pandora	XSuM	XSuM/Pandora	signal	background	overlap	previous chain: XSuM/Pandora, signal, background
vertex in FV	34,486	34,678	1.006	+42	+150	0.863	1.012 -25 +423
nu_score	30,583	30,715	1.004	+60	+72	0.833	1.012 +6 +355
2prong	11,711	11,804	1.008	-15	+108	0.537	1.009 -75 +183
2prong-contained	3,555	3,654	1.028	-36	+135	0.420	1.046 -29 +194
track score	2,689	2,717	1.010	-15	+43	0.435	1.023 -16 +79
vertex distance	1,963	1,966	1.002	+8	-5	0.400	1.008 -24 +40
muon x	1,149	1,151	1.002	-8	+10	0.388	0.966 -48 +9
mu / p PID	947	993	1.049	+25	+21	0.404	0.999 -10 +9

signal / background: XSuM minus Pandora in selected events, primary signal definition. overlap: fraction of the Pandora-selected neutrinos that XSuM also selects (per-event agreement). Statistical precision of the purity at 2prong-contained: 0.008. Pre-2prong background excess: +150 against +423 for the previous chain (vertex residual).

A. Background excess at containment (+135): the slice generator

Where: CC Npi and NC 2prong slices
CC Npi contained fraction of its 2prong slices 0.335 vs 0.290 (256 vs 222 events); NC 2prong +45 events (1,392 vs 1,347) with the containment rate right. Other 0pi and nue: within statistics. The pre-2prong background excess fell from +423 to +150 with the vertex residual.

Cause: the exit <-> multiplicity link of CC + pion slices
In Pandora an exiting muon leaves fewer PFPs: n_pfp = 3 slices have the longest end in the FV 0.418 (0.56 at n_pfp >= 5); generated 0.498 at n_pfp = 3, flat in n_pfp. 2prong is 94% of n_pfp = 3, so the extra contained CC Npi passes the cut. At the slice level (vertex in FV, nu_score > 0.45): P(n_pfp = 3 | exit) - P(n_pfp = 3 | contained) = +0.025 Pandora, +0.011 generated.

Zone 1: true vertex in the FV (30%)
The link runs through the reco-vertex FV cut. Was: a 4-6 cm vertex core (Pandora 0.5 cm) pushed exiting-muon events out of the FV (pass 0.932). Fixed: vertex as the residual to the true vertex, median 0.4 cm; FV pass by exit / contained 0.970 / 0.974 (Pandora 0.983 / 0.965).

Zone 2: dirt neutrinos (61% of these slices)
Reco vertex on the muon line inside the AV; the longest track then runs to the wall. Pandora: P(exit | reco vertex in FV) 0.750, length = path to the wall for 0.51 of exiting tracks. Generated: 0.627 (0.55 before the wall flags), 0.47. The vertex is right; the track given the vertex is not. Open.

Tried and ruled out (same test sample; slice-generator figure, rows 2-4)
Two-stage flow with the counts as numbers (M2s: link closes, count lattice broken, efficiency 0.165); counts as a one-hot class (M2o: counts right, link lost); class from a classifier on the generated stage 1 (M2c: = shipped, link lost); topology sampled after the flow (M2t); 4x topology head; muon-hadron angles as inputs. None reproduces the link at the slice level (0.48-0.52 vs 0.45). Next lever: generate the longest track jointly with the vertex for dirt neutrinos (an exit predictor from the true muon line and the entry wall).

B. Signal mis-assigned at containment: the PFP generator

Where: the 1mu1p0pi neutrinos themselves
Truth-contained signal at 2prong in both chains: contained 0.866 vs 0.984 (-128 of 1,039). Primary signal whose true muon or proton leaves the FV: reconstructed as contained 293 vs 103. Net primary signal -36; truth-contained (notebook) signal -203, notebook purity 0.351 vs 0.418.

Cause: each containment condition is right, their joint is not
Exiting-muon 1p0pi, 2prong in both chains: longest end in FV 0.236 vs 0.224, second end 0.889 vs 0.888, one cathode side 0.674 vs 0.665; but given both ends in the FV, P(vertex and ends on one cathode side) = 0.220 generated vs 0.064 Pandora. A Pandora muon that stops inside the FV after a long path has crossed the cathode; the generated end keeps the vertex side. Contained signal fails the longest-end cut 0.087 (0.007).

What the end point can and cannot be
From generator truth alone the muon end is predictable to 9.9 cm (multiple scattering 9.1); two draws of a perfect surrogate differ by 14.5 cm. Shipped: 11.6 cm per event, lateral scatter 7.8 cm (Pandora 7.9), on the wall 0.363 (0.349). The resolution is at the floor; closed.

What is missing: a correlation, not resolution
Which side of the cathode the end lies on is set by the vertex x, the direction and the length; the set flow smears it. Ruled out: end-point bend fix (M3b), end as a direct target + particle attention (M3e), 2 / 4 attention layers (no change in the FV / wall rates). Geant4 end-point inputs (M3g, M3w) help but are not admissible.

Next lever
Generate the track end relative to the cathode: a per-track flag 'end on the other side of the cathode' (and the crossing point) as a target of the set flow, like the wall flags; or the end x as the signed distance to the cathode. The hybrid (Pandora slices + generated PFPs) shows the same joint loss (P(side | both ends) 0.205 vs 0.086), so the fix is in the PFP generator, not upstream. Expected gain: the notebook purity (0.351 -> 0.418) and the -36 primary signal at containment.

Caveats and smaller items
Splits: the file assignment is a seeded shuffle of the file list; the datasets' test files were re-derived from the generators' training table on 2026-11-09 (before, 71% of them were training files; the closure moved by 0.01). The cardinality head (matched slices per neutrino) is split on the card table and saw 90% of the held-out files; next training round: split by a hash of the file id.
Range momenta are generated as independent numbers (the range peak of stopping pions is smeared, 0.17 vs 0.22); the induction-plane chi2, unfitted PFPs, truth-match columns and cosmic-only slices are not in the files.
Per-column W1 of the slice block: median 0.016 of the transformed range; the largest values are the cohpi end points of the two-track block (5% of slices), measured before the end-point clip (in the dataset: on the wall 0.433 vs 0.429, outside the TPC 0.000 vs 0.001).
Files: reports/cc1p_containment/README.md (all studies), accept_m2vw.txt, accept_m3w0.txt, diag_m2vw_summary.txt, topo_m2vw.txt, m2_purity_m2vw.txt, link_m2vw.txt, vtx_dir_m2vw.txt, vtx_dirt_m2vw.txt, contain_joint_m2vw_noclip.txt.